Exercise 10

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Introduction To Ocean Modelling

Marcello Vichi

I tested the model by changing three parameters and tracking the Courant number throughout, taking a lower Courant number as the marker of a better-resolved solution.

First, I varied the advection speed at fixed resolution (dx). Lowering the speed lowered the Courant number and reduced distortion, so I initially settled near 0.15 m/s to later refine my choice.

I then varied dx at fixed advection speed. This exposed the model's stability limit: at dx = 50 m the Courant number reached 4 and the simulation broke down completely, it became unstable once the Courant number became large. I settled on dx = 150 m, though with some scepticism, since the degree of distortion between the analytical and the numerical remains.

Combining a moderate dx (150 m) with a low speed (0.20 m/s, Courant ≈ 0.27) resulted in minimised distortion.Finally I varied σ. A wider pulse is better as it holds its shape, which led me to σ = 2750 m.

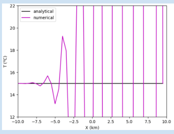
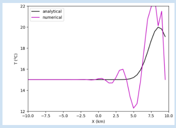
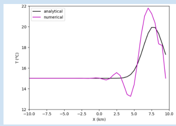
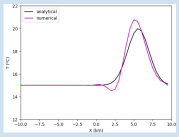
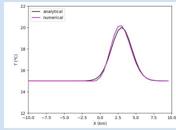
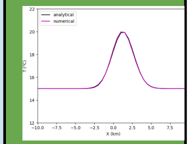
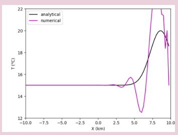
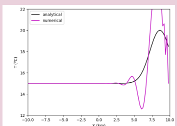
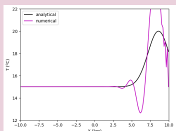
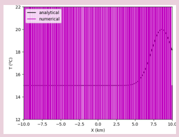
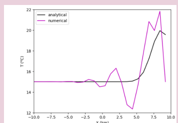
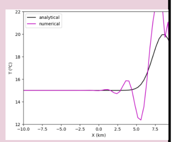
Selected configuration: σ = 2750 m, dx = 150 m, c = 0.20 m/s (Courant ≈ 0.27) because it minimises smearing, and keeps it clear of the faulting boundary.

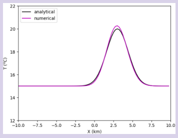
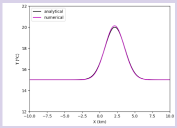
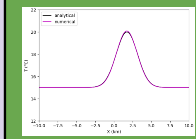
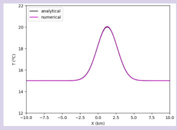
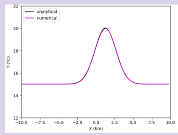
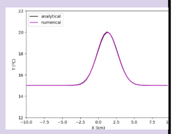
I asked Claude AI: ‘Why does my moderate dx (150m) and low speed of 0.20m/s not result in a distorted seen in the dx test runs:

Claude AI Response: At a low speed of 0.20m/s, the pulse travels only about 1.7 km over the 0.1-day run (0.2 m/s × 8640 s), while the downstream boundary sits 10 km from the pulse's starting point at the centre. The pulse therefore never gets close to the clamped edge, so the boundary stays at background temperature throughout and the fixed-value condition never faults. The distortion only arises at high speeds — for example at 2 m/s the pulse travels 17.3 km and collides with the boundary partway through the run.

Table 1: Testing Model Outputs through changing advection speed, resolution and sigma.

Advection speed = 2m/s	Advection speed = 1m/s	Advection speed = 0.90m/s	Advection speed = 0.65m/s	Advection speed = 0.35m/s	Advection speed = 0.15m/s

					
Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 2 m/s Courant number = 0.8	Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 1 m/s Courant number = 0.4	Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 0.9 m/s Courant number = 0.36	Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 0.65 m/s Courant number = 0.26	Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 0.35 m/s Courant number = 0.14	Grid steps : 40 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.5 m/s Physical speed = 0.15 m/s Courant number = 0.06
dx = 250m	**Dx = 200m**	**dx= 150m**	**dx= 50m**	**Dx = 800m**	**dx= 400m**
					
Grid steps : 80 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 1.25 m/s Physical speed = 1 m/s Courant number = 0.8	Grid steps : 100 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 1.0 m/s Physical speed = 1 m/s Courant	Grid steps : 134 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 0.75 m/s Physical speed = 1 m/s Courant	Grid steps : 400 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 0.25 m/s Physical speed = 1 m/s Courant	Grid steps : 25 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 4.0 m/s Physical speed = 1 m/s Courant number = 0.25	Grid steps : 50 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.0 m/s Physical speed = 1 m/s

	number = 1.0	number = 1.3333333333333333	number = 4.0		Courant number = 0.5
Advection speed = 0.35 m/s **Dx = 250m**	**Advection speed = 0.25m/s** **Dx = 190m**	**Advection speed = 0.20m/s** **Dx = 150m**	**Advection speed = 0.15m/s** **Dx = 150m**	**Advection speed = 0.15m/s** **Dx = 250m**	**Advection speed = 0.15m/s** **Dx = 400m**
					
Grid steps : 80 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 1.25 m/s Physical speed = 0.35 m/s Courant number = 0.28	Grid steps : 106 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 0.95 m/s Physical speed = 0.25 m/s Courant number = 0.2631578947368421	Grid steps : 134 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 0.75 m/s Physical speed = 0.2 m/s Courant number = 0.26666666666666666	Grid steps : 134 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 0.75 m/s Physical speed = 0.15 m/s Courant number = 0.2	Grid steps : 80 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 1.25 m/s Physical speed = 0.15 m/s Courant number = 0.12	Grid steps : 50 Number of time steps : 44 Timestep = 200 s Max numerical speed DX/DT = 2.0 m/s Physical speed = 0.15 m/s Courant number = 0.075
Σ = 1500m **Advection speed = 0.25m/s** **Dx = 150m**	**Σ = 2500m** **Advection speed = 0.25m/s** **Dx = 150m**	**Σ = 2750m** **Advection speed = 0.25m/s** **Dx = 150m**